Just call it a "p-value"!

(not a hypothesis probability,
not a false alarm probability)

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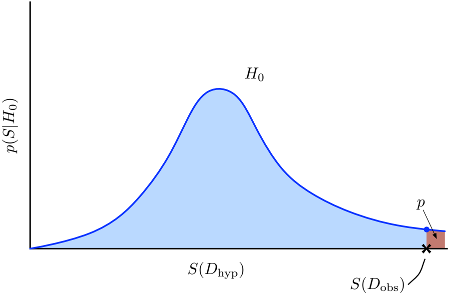
CASt Summer School — June 2026

p-value for test statistic S, null H_0

$$\begin{aligned} p &= P(S(D_{\rm hyp}) > S(D_{\rm obs})|H_0) \\ &= P(\{D_{\rm hyp} : S(D_{\rm hyp}) > S(D_{\rm obs})\}|H_0) \end{aligned}$$



Introduced by Fisher, who simply called it "p" (whence "p-value").

Smaller p-values indicate stronger evidence against H_0.

Astronomers call this the *significance level* or (sometimes) *false-alarm probability*. Statisticians don't—for good reason!

The p-value controversy, ca. 2016

- Context: Reproducibility crisis in science
  - *PLOS Medicine* 2005: "Why most published research findings are false" by John Ioannidis
  - 2010: *The New Yorker* — "The truth wears off: Is there something wrong with the scientific method?"
  - *PLOS Biology* 2015: $\approx$ \$28B/yr on research that cannot be replicated
- Re: *statistical* reproducibility: Journals reassess $p < 0.05$
- Culminates in ASA's Statement on p-values (March 2016)

ASA statement on statistical significance and p-values

- ***P*-values do not measure the probability that the studied hypothesis is true, or the probability that the data were produced by random chance alone.**
- **Scientific conclusions and business or policy decisions should not be based only on whether a *p*-value passes a specific threshold.**
- **By itself, a *p*-value does not provide a good measure of evidence regarding a model or hypothesis.**
- ...

Impact on behavioral sciences

Title: Redefine Statistical Significance

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One Sentence Summary: We propose to change the default *P*-value threshold for statistical significance for claims of new discoveries from 0.05 to 0.005.

Nature Human Behavior (2018), DOI:10.1038/s41562-017-0189-z

Agenda

- p-values: What they are, and are not
- Significance tests: Neyman-Pearson testing vs. p-values
- Getting FAPs from p-values— A Monte Carlo experiment
- Hey, those FAPs are like Bayesian posterior odds!

An old misunderstanding: p as a probability for the null hypothesis

The Astrophysical Journal, 782:100 (14pp), 2014 February 20 doi:10.1088/0004-637X/782/2/100

THE ABSOLUTE MAGNITUDE DISTRIBUTION OF KUIPER BELT OBJECTS

ABSTRACT

Here we measure the absolute magnitude distributions (*H*-distribution) of the dynamically excited and quiescent (hot and cold) Kuiper Belt objects (KBOs), and test if they share the same *H*-distribution as the Jupiter Trojans. From

> *"The Kolmogorov-Smirnov test reveals that the probability that the Trojans and cold KBOs share the same parent H-distribution is less than 1 in 1000."*

p-values are probabilities for *data*, not hypotheses—only Bayesian methods can give probabilities for hypotheses.

Resurrected in different language: *p*-values and the FAP fallacy

FAP = False Alarm Probability

"This detection has a signal-to-noise ratio of 4.1 with an empirically estimated upper limit on false alarm probability of 1.0%."

"...the false alarm probability for this signal is rather high at a few percent."

"This signal has a false alarm probability of < 4 % and is consistent with a planet of minimum mass 2.2 $M_\odot$..."

"We find a false-alarm probability $< 10^{-4}$ that the RV oscillations attributed to CoRoT-7b and CoRoT-7c are spurious effects of noise and activity."

"The signal was observed with a matched-filter signal-to-noise ratio of 24 and a false alarm rate estimated to be less than 1 event per 203,000 years, equivalent to a significance greater than 5.1 σ."

All of these statements incorrectly describe the weight of evidence for a detection, and almost certainly greatly exaggerate the weight of the evidence.

What's wrong?

"This *signal, with* $S(D_{\rm obs}) = X$, *has a* FAP *of* $p\ldots$"

$$p = P(\{D_{\rm hyp} : S(D_{\rm hyp}) \geq S(D_{\rm obs})\}|H_0)$$

The statement is inconsistent with the calculation producing p in two ways:

Probability ... **given** $\boldsymbol{H_0}$

p is computed assuming that *H_0 always operates*

Every alarm is false—i.e., with FAP$= 1$—in this "world" of replications (the context of the calculation producing p)

For any signal to have FAP$\neq 1$, *alternatives* to the null must sometimes act; the FAP will depend on how often they do (and what they are)

Probability... including **worse** *departures from null predictions*

p refers, not specifically to $D_{\rm obs}$, but to a *set* including *more extreme data* — it's a property of that whole ensemble, not of $D_{\rm obs}$ by itself

$D_{\rm obs}$ bounds this set on the side of *weakest* evidence vs. H_0

Harold Jeffreys, addressing an audience of statisticians (1980):

> I have always considered the arguments for the use of P absurd. They amount to saying that a hypothesis that may or may not be true is rejected because a greater departure from the [observed] trial was improbable; that is, that it has not predicted something that has not happened. As an argument astronomer's experience [requiring several standard deviations departures] is far better.

What a p-value really means

In the voice of Don LaFontaine or Lake Bell:

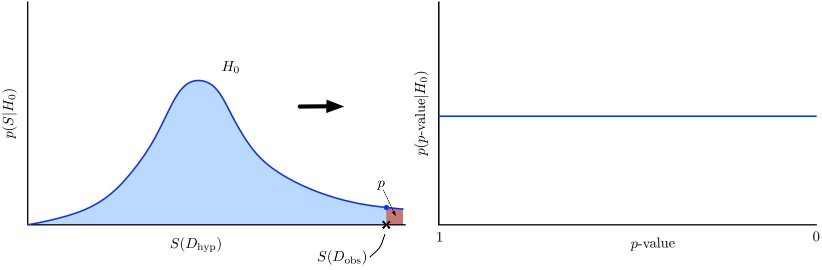
In a world. . . *with absolutely no sources, with a threshold set so we wrongly claim to detect sources* $100 \times p\%$ *of the time, this data would wrongly be considered a detection—and it would be the data providing the* **weakest** *evidence for a source in that world.*

Who wants to say *that*?! Whence "p-value."

p's one intuitively accessible property

Under the null, for any $X \in [0, 1]$, the fraction of time $p > X$ is. . . simply X!

Think of p as an alternative test statistic—a nonlinear mapping of $S(D)$ to the unit interval—that has a *uniform distribution under the null*



p is a (particular) surprise-ordered relabeling of the data, with a $U(0, 1)$ PDF, and a linearly rising CDF

Surprise isn't enough

The rarity of data "like" $D_{\rm obs}$ under H_0 is evidence against H_0 only if *plausible alternatives* make $D_{\rm obs}$ *less* surprising.

Expand the "world" of the p-value calculation:

- Let an **alternative**, H_1, sometimes operate, with probability π_1 (with null prevalence $\pi_0 = 1 - \pi_1$)
- Compare the rates for getting the **observed** p-value under H_0 and H_1 (*not* "observed or smaller p-value")
- Equivalently: Compare the rates for getting $S(D) = S(D_{\rm obs})$ under H_0 and H_1 (*not* "observed or larger $S(D)$")

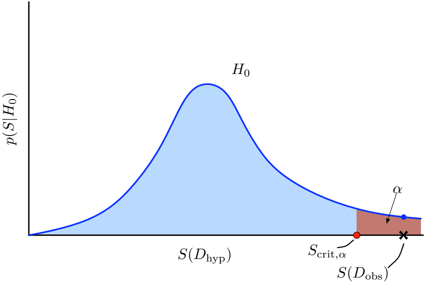
This *conditional frequentist* approach produces genuine FAPs.

(See below for a Monte Carlo experiment demonstrating this.)

Significance Testing and p-values

Neyman-Pearson testing

- Specify simple null hypothesis H_0 such that rejecting it implies an interesting effect is present
- Devise statistic $S(D)$ measuring departure from null
- Divide sample space into probable and improbable parts (for H_0); $p(\text{improbable}|H_0) = \alpha$ (Type I error rate), with α specified a priori
- If $S(D_{\rm obs})$ lies in improbable region, reject H_0; otherwise accept it
- Report: "H_0 was rejected (or not) with a procedure with false-alarm frequency α" (aka Type 1 error rate, test size)



Neyman and Pearson devised this approach guided by Neyman's *frequentist principle*:

> *In repeated practical use of a statistical procedure, the long-run average actual error should not be greater than (and ideally should equal) the long-run average reported error. (Berger 2003)*

A *confidence region* is an example of a familiar procedure satisfying the frequentist principle.

They insisted that one also specify an alternative, and find the error rate for falsely rejecting it (Type II error).

Note that α is specified a priori, and is a property of the test (viewed as a procedure applied across an ensemble), not a property of a particular outcome.

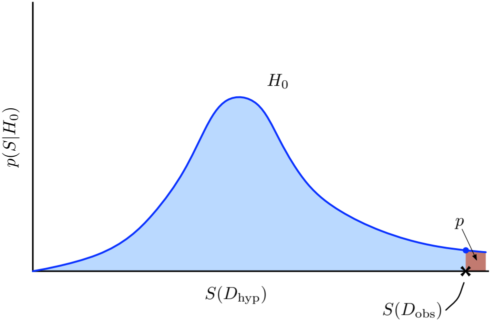
Fisher's p-value testing

Fisher felt reporting a rejection frequency of α no matter where $S(D_{\mathrm{obs}})$ lies in the rejection region does not accurately communicate the strength of evidence against H_0.

He advocated reporting the *p-value*:

$$p = P(S(D) > S(D_{\mathrm{obs}})|H_0)$$

Smaller p-values indicate stronger evidence against H_0.



p-value complications

Fisherian testing does not have the straightforward frequentist properties of NP testing, but everyone uses it anyway.

E.g., rejections of H_0 with p-value$= 0.05$ are *not* "wrong 5% of the time under the null" or "with 5% false-alarm probability." They are wrong *100% of the time* under the null. To quantify the conditional error rate (i.e., the error rate among datasets with the same p-value), you *must* say something about the alternative.

Even NP tests have unpleasant frequentist properties; e.g., the strength of the evidence against the null (e.g., quantified by a conditional false alarm rate) for a fixed-α test grows weaker as N increases. NP themselves advocated decreasing α with N, but there are no general rules for this.

False alarm rates

Berger (2003) discusses the relationship between p-values, false alarm rates, and Bayesian posterior probabilities (or odds and Bayes factors).

In simple settings where one can easily bound false alarm rates, he shows the p-value significantly underestimates the false alarm rate among datasets sharing a given p-value.

This gives insight into why we've come to consider apparently small p-values—like "2σ" ($p \approx 0.05$) or "3σ" ($p \approx 0.003$)—to represent only weak evidence against the null. Typically, datasets with such p-values are not much more probable under alternatives than under the null.

He also shows that a "conditional frequentist" calculation of the false alarm rate in some simple settings amounts to computation of a Bayes factor. (See example below.)

Entries to the literature (not up to date)

- "402 Citations Questioning the Indiscriminate Use of Null Hypothesis Significance Tests in Observational Studies" (Thompson 2001) [web site]
- *The significance test controversy: a reader* (ed. Morrison & Henkel 1970, 2006) [Google Books]
- "Could Fisher, Jeffreys and Neyman Have Agreed on Testing?" (Berger 2003 with discussion; 2001 Fisher award Lecture), *Statistical Science*, **18**, 1–32 [journal site—highly recommended!]
- "Odds Are, It's Wrong: Science fails to face the shortcomings of statistics" (By Tom Siegfried 2010) [*Science News*, March 2010]
- "Scientific method: Statistical errors" (By Regina Nuzzo 2014) [*Nature* news feature, Feb 2014]
- "The ASA's statement on p-values: context, process, and purpose" [*The American Statistician*, March 2016]

A Monte Carlo experiment (Berger 2003)

Consider measurements of μ with Gaussian noise, σ known:

Model: $x_i = \mu + \epsilon_i$, ($i = 1$ to n) $\qquad \epsilon_i \sim N(0, \sigma^2)$

Null hypothesis, H_0: $\mu = \mu_0 = 0$

Alternatives (composite H_1): $\mu \sim N(0, 4\sigma^2)$

Hypothesis prior: $p(H_0) = p(H_1) = 0.5$

Test statistic:

$$t(x) = \frac{|\bar{x}|}{\sigma/\sqrt{n}}$$

p-value:

$$\begin{aligned} p(t|H_0) &= \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \\ p\text{-value} &= P(t \geq t_{\rm obs}) \end{aligned}$$

t	p-value
1	0.317
2	0.046
3	0.003

$p = .05 \rightarrow$ "significant"

$p = .01 \rightarrow$ "highly significant"

Collect the p-values from a large number of tests in situations where the truth eventually becomes known, and determine how often H_0 is true at various p-value levels.

- Suppose that, overall, H_0 was true about half of the time.
- Focus on the subset with $t \approx 2$ (say, $[1.95, 2.05]$ so $p \in [.04, .05]$, so that H_0 was rejected at the 0.05 level.
- Find out how many times in that subset H_0 turned out to be true.
- Do the same for other significance levels.

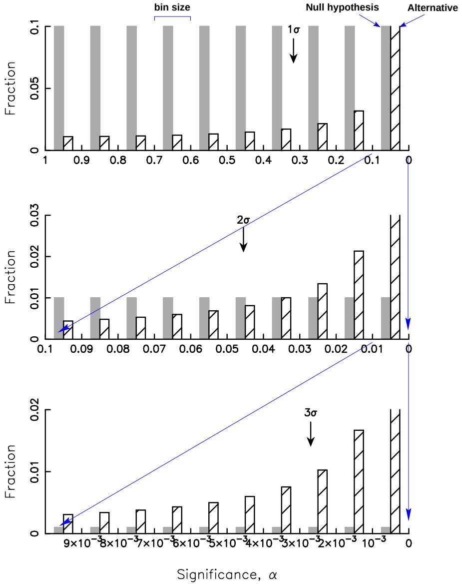
Monte Carlo implementation details

- Choose $\mu = 0$ **or** $\mu \sim N(0, 4\sigma^2)$ with a fair coin flip*
- Simulate n data, $x_i \sim N(\mu, \sigma^2)$ (use $n = 20$, 200, 2000)
- Calculate $z_{\rm obs} = \frac{|\bar{x}|}{\sigma/\sqrt{n}}$ and $p(z_{\rm obs}) = P(z > z_{\rm obs}|\mu = 0)$
- Bin $p(z)$ separately for each hypothesis; repeat

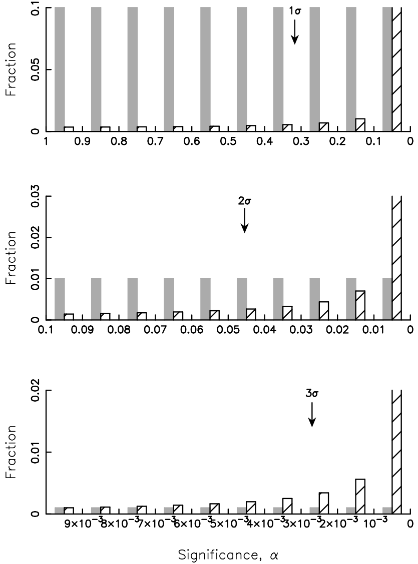
Compare how often the two hypotheses produce data with a 1–, 2–, or 3–σ effect $\rightarrow$ *conditional error probabilities* (real FAPs!)

*A neutral assumption that gives alternatives a "fair" chance and may *over*estimate the evidence against H_0 in real settings where the null is more prevalent

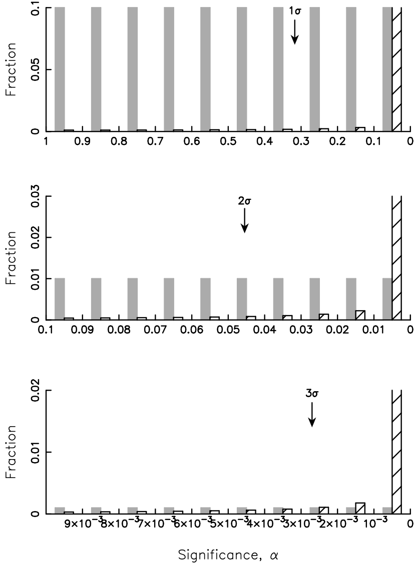
Significance Level Frequencies, $n = 20$



Significance Level Frequencies, $n = 200$



Significance Level Frequencies, $n = 2000$



What about another μ prior?

- For data sets with H_0 rejected at $p \approx 0.05$, H_0 will be true *at least* 23% of the time (and typically close to 50%). (Edwards et al. 1963; Berger and Selke 1987)
- At $p \approx 0.01$, H_0 will be true *at least* 7% of the time (and typically close to 15%).

What about a different "true" null frequency?

- If the null is initially true 90% of the time (as has been estimated in some disciplines), for data producing $p \approx 0.05$, the null is true at least 72% of the time, and typically over 90%.

In addition . . .

- At a fixed p, the proportion of the time H_0 is falsely rejected grows as $\sqrt{n}$. (Jeffreys 1939; Lindley 1957)
- Similar results hold generically; e.g., for χ^2. (Delampady & Berger 1990)

A p-value is not an easily interpretable measure of the weight of evidence against the null.

- It does not measure how often the null will be wrongly rejected among similar data sets
- A naive false alarm interpretation typically overestimates the evidence
- For fixed p-value, the weight of the evidence decreases with increasing sample size

Conditional error rates (actual FAPs!) and Bayesian posterior odds

Bayes's theorem comparing two hypotheses $\rightarrow$ posterior odds:

$$
\begin{aligned}
O_{10} &\equiv \frac{p(H_1|D)}{p(H_0|D)} \\
&= \frac{p(H_1)}{p(H_0)} \times \frac{p(D|H_1)}{p(D|H_0)}
\end{aligned}
$$

Here $D = \{x_i\}$, and the *Bayes factor* is:

$$
B \equiv \frac{p(\{x_i\}|H_1)}{p(\{x_i\}|H_0)}
$$

For this model, it's straightforward to show:

$$B \equiv \frac{p(\{x_i\}|H_1)}{p(\{x_i\}|H_0)} = \frac{p(p_{\rm obs}|H_1)}{p(p_{\rm obs}|H_0)}$$

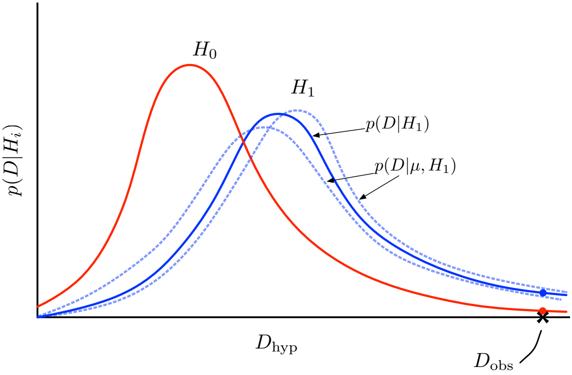
$\rightarrow B$ here is just the ratio of adjacent bin heights (solid vs. hatched) calculated in the Monte Carlo, comparing false and true alarm rates!*

Bayes factors can give you FAPs—directly if the equal-prior assumption is reasonable, or adjusted for the hypothesis prior if another choice is relevant.

*This happens because there is a sufficient statistic, and the test statistic uses it; i.e., all of the information in the data is being used in this simple example, not just the information in an incomplete summary statistic, as may generally be the case.

For *compound* hypotheses (H_1 here), the *marginal likelihood* accounts for parameter uncertainty that is ignored by *p*-values (which typically set parameters equal to best-fit values):

$$p(D|H_i) = \int d\theta_i \; p(\theta_i) \; p(D|\theta_i, H_i)$$



Also, the marginal likelihood uses *all* of the data, not just the value of a test statistic: in general $p(D|H_i) \neq p(S(D)|H_i)$

Why is the p-value a poor measure of the weight of evidence?

- We should be *comparing hypotheses*, not trying to identify rare/surprising events—an observation surprising under the null motivates rejection only if it is not surprising under reasonable alternatives
- Comparison should use the *actual data*, not merely membership of the data in some larger set. A p-value conditions on incomplete information.

Harold Jeffreys, addressing an audience of statisticians (1980):

> For n from about 10 to 500 the usual result is that *[the Bayes factor]* $K = 1$ when $(a - \alpha_0)/s_\alpha$ is about 2. . . not far from the rough rule long known to astronomers, i.e. that differences up to twice the standard error usually disappear when more or better observations become available. . . *[nowadays we'd want 4σ or 5σ so the detection is unambiguous, with $K \gg 1$]*....
>
> I have always considered the arguments for the use of P absurd. They amount to saying that a hypothesis that may or may not be true is rejected because a greater departure from the [observed] trial was improbable; that is, that it has not predicted something that has not happened. As an argument astronomer's experience is far better.